



B.Tech. Degree VI Semester Regular/Supplementary Examination

May 2024

ME 19-205-0606 (IE) ADDITIVE MANUFACTURING

(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

- CO1: Understand the concept of AM.
 CO2: Study various AM technologies using the basics of computer numerical control.
 CO3: Study the modeling of AM processes.
 CO4: Study the application of various fields.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate.
 L6 – Create

PO – Programme Outcome

PART A

(Answer *ALL* questions)

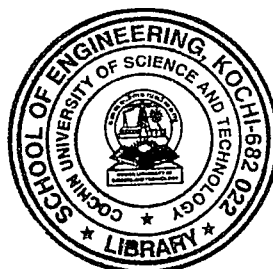
		(8 × 3 = 24)	Marks	BL	CO	PO
I.	(a)	Define a prototype and enumerate its roles.	3	L1	1	1
	(b)	What is additive manufacturing? List out its advantages.	3	L2	1	1,2
	(c)	“Anisotropy is an important problem for AM”.Explain.	3	L2	2	1,2
	(d)	Explain about functionally graded materials.	3	L1	2	1,2
	(e)	What are the differences between selective laser sintering and selective laser melting?	3	L1	4	1,2
	(f)	What are the implications of vacuum in electron beam powder bed fusion?	3	L2	4	1,2
	(g)	Enumerate various numerical methods for deposition modeling.	3	L1	3	1,2
	(h)	In the parlance of modelling of free surface flow how, do you distinguish between Lagrangian methods and Eulerian methods.	3	L2	3	1,2

PART B

(4 × 12 = 48)

II.	(a)	Explain the Generic AM Process.	7	L2	1	1,2,3
	(b)	What is indirect Rapid Tooling?	5	L2	1	1,2
OR						
III.	(a)	Explain about AM application levels with sketches.	7	L1	1	1,2
	(b)	Explain the basic theory of reverse engineering.	5	L2	1	1

(P.T.O.)



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		Marks	BL	CO	PO
IV.	(a) Discuss about various materials used in AM.	7	L1	2	1,2,
	(b) Compare the anisotropic behaviour of materials in various AM technologies.	5	L2	2	1,2,
OR					
V.	(a) Explain about various factors affecting solidification in AM.	7	L2	2	1,2,
	(b) What kinds of plastics can be processed by AM? Plot them in the plastic triangle.	5	L1	2	1,2,
VI.	(a) Explain laser powder bed fusion process with a neat sketch.	6	L1	4	1,2,
	(b) Explain various components needed to realize electron beam powder bed fusion.	6	L1	4	1,2,
OR					
VII.	(a) Explain the working principles of fused deposition modeling.	7	L1	4	1,2,
	(b) Explain about process planning in AM..	5	L1	4	1,2,
VIII.	Explain the governing equations used in the mathematical modeling of laser metal deposition process	12	L2	4	1,2,
OR					
IX.	Explain thermo-hydrodynamic simulation in Additive Manufacturing Processes.	12	L2	4	1,2,

Blooms's Taxonomy Level's

L1 – 55%, L2 – 45%.
